Problem I
One Road Down

Time limit: 2 seconds

A city has N junctions connected by M two-way roads. Each road has a fixed travel time in minutes, which is the same in both directions.

The ambulance station is at junction S, and the main hospital is at junction T. Ambulances always take the fastest available route from S to T.

The roads are old, so the city closes one road each night for repairs. Before deciding the repair schedule, the planners want to know how each closure would affect the ambulance service.

For each road independently, find the fastest travel time from S to T when **only that road is closed**. All other roads remain open.

Input

The first line contains four integers N, M, S and T ($2 \le N \le 100000$, $1 \le M \le 200000$, $1 \le S, T \le N$, $S \ne T$), the number of junctions, the number of roads, the ambulance station's junction and the hospital's junction.

Junctions are numbered from 1 to N.

Each of the next M lines contains three integers U, V and W ($1 \le U, V \le N$, $U \ne V$, $1 \le W \le 10^9$). They describe a two-way road between junctions U and V with a travel time of W minutes.

Roads are numbered from 1 to M in input order. There may be several roads between the same pair of junctions. These are separate roads, and closing one does not close the others.

When all roads are open, every junction is reachable from every other junction.

Output

Print M lines.

On line i, print the fastest travel time in minutes from S to T when road i is closed. Print -1 if the hospital cannot be reached during that closure.

Sample Input 1	Sample Output 1
7 9 1 7 1 2 2 2 3 2 3 6 2 1 4 3 4 3 2 2 5 4 5 6 3 4 5 6 6 7 1	8 8 10 7 7 7 7 7 -1